

Centenary Celebrated Sharnbasveshwar Vidya Vardhaka Sangha's



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SHARNBASVA



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UNIVERSITY



A State Private University approved by Govt. of Karnataka vide Notification No. ED 144 URC 2016 dated 29-07-2017
Recognised by UGC under Section 2f vide No. F.8-29/2017 (CPP-I/PU), dated 20-12-2017 & AICTE, CoA, PCI New Delhi

CERTIFICATE

Certified that Project work entitled **"INTELLIHELMET: ADVANCED IOT-ENABLED ALCOHOL DETECTION SYSTEM FOR ENHANCED ROAD SAFETY"** carried out by **SOMYA R MITEKAR (USN: SW22SCS008)** in partial fulfillment of the requirements for the award of the degree M.Tech in **COMPUTER SCIENCE & ENGINEERING**, Sharnbasva University, Kalaburagi during the academic year 2023-24. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The Project Report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of guide

Prof. CHETHANA S PATIL

Signature of Chairperson

CHAIRMAN
Computer Science Engineering
Faculty of Engineering & Technology
(Exclusively For Women)
Sharnbasva University, Kalaburagi

Signature of Dean

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Names of Examiners:

- 1.
- 2.

Signature with Date

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DECLARATION

I, **SOUMYA R MITEKAR** USN: **SW22SCS008**, student of Sharnbasva University Kalaburagi, hereby declare that the project title **"INTELLIHELMET: ADVANCED IOT-ENABLED ALCOHOL DETECTION SYSTEM FOR ENHANCED ROAD SAFETY"** embodies the report of my project carried out independently by me during 4th semester of Master of Technology in **COMPUTER SCIENCE AND ENGINEERING** at Sharnbasva University under the supervision and guidance of **Prof. CHETHANA S PATIL**. Thus work has been submitted as partial fulfillment of the requirement for the award of the degree of **Master of Technology in Computer Science And Engineering by Sharnbasva** University Kalaburagi, during the academic year 2023-2024

I have not submitted the matter that embodies to any other university or institution for the reward of any other degree

SOUMYA R MITEKAR
(USN: SW22SCS008)



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ABSTRACT

At this time, accidents are major concern for all individuals. Frequency of accidents is rising, prompting measures to be taken to prevent or lessen their impact. Regulations of road are routinely disregarded and disregarded by people in this planet. Plus, people always fight back when they feel pressured. So, taking new tack, we offer protection via a smart helmet's opulent & sophisticated features. Smart Helmet is project that utilises microcontrollers from AVR family. It is a headgear equipped with intelligent technologies that enhance the driving experience while improving safety. A synchronised effort by two modules mounted on bike & helmet will guarantee rider is wearing the protective headgear. wireless connection among bike circuit & helmet is accomplished by use of radio frequency module. To avoid situations when someone is drunk and driving, ALCHO-LOCK mechanism is used. Our circuit incorporates accelerometer for accident detection and a GSM module to automatically notify one individual along with appropriate official which person was involved in accident. Additionally, we use fog sensor to enhance visibility in foggy or smoggy conditions. It is believed that the initiative would make roads safer and reducing number of accidents, particularly those that kill motorcyclists.

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